Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name:** Andon Agent
b. **Team leader name:** Likhith M
c. **Problem Statement:** **Manufacturing: Intelligent Operations & Industrial Efficiency**

When an automotive assembly line stops, the alarm often fires downstream of the real cause. Find a verified root cause fast and hand a human a safety-gated work order, without touching a machine.

Brief about the idea

$2.3M

lost per hour when a large automotive plant stops

Siemens, The True Cost of Downtime 2024, p.2

The problem

The andon alarm fires where the symptom shows (a weld cell starved of air), not where the cause is (a plant-air header upstream). Engineers stitch telemetry, alarms, shift notes and manuals together by hand while the line waits.

Andon Agent investigates a stopped line like a reliability engineer, then hands a human a verified root cause and a safety-gated draft work order.

1 Investigate

Commander plans competing checks; agents run change points, spectra and machine-sound analysis on telemetry, read alarms and shift notes, cite the manual.

→

2 Verify

Deterministic code (no LLM) re-runs every number, timestamp, quote and causal claim. Rejected claims are struck through, never hidden.

→

3 Approve

Draft work order with lockout/tagout steps quoted verbatim with page numbers. A named human approves; nothing touches a machine.

0.98

top-1 root cause, 60 synthetic scenarios

0.76

top-1, 17 real-signal scenarios

1 / 586

claims code could not back (vs 54 / 645)

≈ $0.11

per diagnosis, 81 s median

Eval run final_v2 (Gemini 3.8 Flash judge); 95% Wilson intervals on slide 13. Real public signals mapped onto a synthetic line, not real plant data.

Who it serves

L Line technician

PAIN TODAY

Sent to the station that alarmed; safety steps live in a thick English manual.

WHAT ANDON AGENT GIVES

/tech phone view: one card at a time, big verbatim LOTO checklist, voice note → shift note, work order in Hindi, Tamil, Bahasa Indonesia or Vietnamese beside the binding English text.

M Maintenance supervisor

PAIN TODAY

Must approve repair work without seeing why the cause was chosen.

WHAT ANDON AGENT GIVES

Ranked root cause with verified evidence, draft work order PENDING_APPROVAL, approval recorded with the approver's name in a mock CMMS.

R Reliability engineer

PAIN TODAY

Stitches historian exports, alarm logs, shift notes and manuals by hand.

WHAT ANDON AGENT GIVES

Evidence drawer charts the telemetry behind every claim; change points, spectra, audio; upload your own CSV and investigate it.

P Plant manager

PAIN TODAY

Every stopped minute is worth about $38,333 (Siemens $2.3M/h ÷ 60).

WHAT ANDON AGENT GIVES

Cost per run on screen (≈ $0.11), append-only audit of every model call and human decision, no physical writes.

Opportunities

how it differs · how it solves · USP

	Dashboards	Rule engines	RAG chatbot	Andon Agent
Runs analytics on signals and machine sound	Charts only	No	No	**Yes**
Forms competing hypotheses, ranks causes	No	Fixed rules	Partly	**Yes**
Code re-checks every claim before display	n/a	n/a	No	**Yes**
Safety steps quoted verbatim with page	No	No	Paraphrases	**Yes**
Drafts action; a human approves	No	Alerts only	No	**Yes**

“Isn't this just RAG?” No.

RAG retrieves text and paraphrases it. Andon Agent runs analytics on telemetry and audio, forms competing hypotheses, and every claim cites evidence ids that deterministic code re-runs before a human acts. Measured against one long prompt given the same evidence: **1 of 586 claims unsupported vs 54 of 645.**

HOW IT SOLVES THE PROBLEM

- Works upstream from the alarming station to the real cause
- Fuses telemetry, alarms, shift notes, manuals and machine audio
- Abstains (insufficient evidence) instead of guessing
- Ends in a draft work order, never an actuator

USP

The model never grades itself.

Verifier stress test: 2,778 injected faults, 97.1% caught; without the verifier 0%. Safety text is inserted by code, verbatim.

List of features offered by the solution

all built and live

1 Incident console

Line mimic of synthetic Line L1 with utilities rail; stopped station and root cause marked.

2 Live agent trace

Every plan, tool call and result streams in (SSE), with provenance chips on real signals.

3 7-step stepper

Stop → Plan → Evidence → Hypotheses → Verify → Work order → You.

4 Evidence drawer

Click any evidence id: the telemetry window behind the claim is charted (real signal where marked).

5 Deterministic verifier

Code re-checks each claim; rejected ones are struck through with the measured value.

6 Root cause + actions

Ranked causes citing evidence ids; accept, edit or reject; downtime cost estimate.

7 Safety-gated work order

LOTO steps verbatim with page numbers; approver name; human approval → mock CMMS.

8 Multilingual work order

Hindi, Tamil, Bahasa Indonesia, Vietnamese machine translation beside untouched English safety text.

9 Bring your own CSV

Gemini Flash-Lite proposes a column → tag mapping; a human confirms; a normal investigation runs.

10 Technician view /tech

Phone-first cards, big safety checklist, voice note → structured shift note marked untrusted.

11 Replay + per-run cost

Recorded runs replay free for anyone; live runs show tokens and $ per diagnosis.

12 /eval page

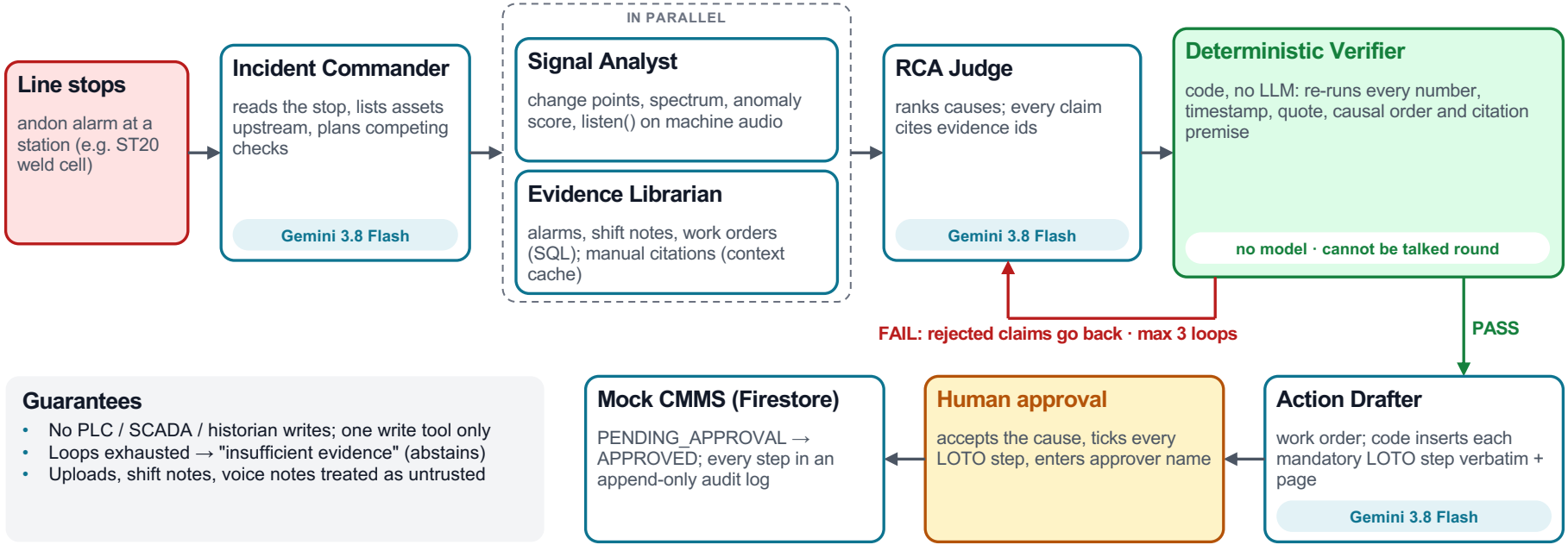
Committed benchmark results with intervals, inside the app.

Process flow diagram

one stopped line, end to end

1 Stop → 2 Plan → 3 Evidence → 4 Hypotheses → 5 Verify → 6 Work order → 7 You

The console's investigation stepper, step by step

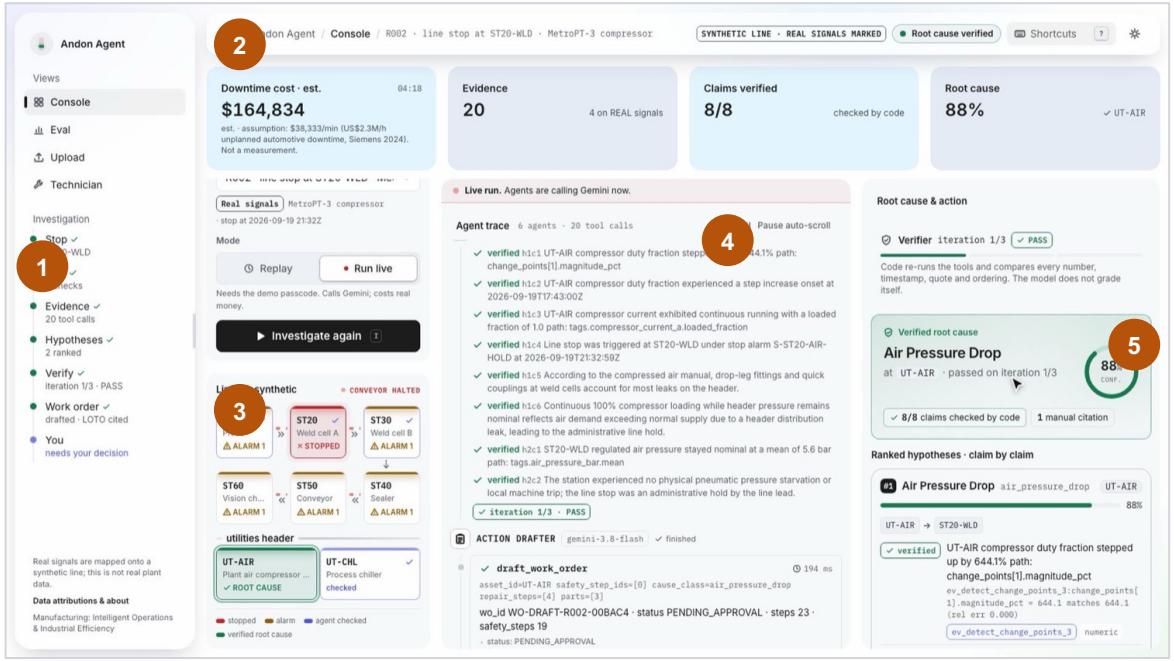


Guarantees

- No PLC / SCADA / historian writes; one write tool only
- Loops exhausted → "insufficient evidence" (abstains)
- Uploads, shift notes, voice notes treated as untrusted

Wireframes / mock

real deployed screens, annotated · live run, 17 Oct 2026

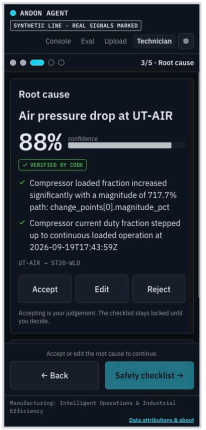


Live run, 17 Oct 2026 · R002, real MetroPT-3 signals on the synthetic line · 20 tool calls · 8/8 verified · 88% at UT-AIR

1. **Investigation stepper**
   Stop → Plan → Evidence → Hypotheses → Verify → Work order → You
2. **Status strip**
   downtime estimate (labelled assumption), claims verified, confidence
3. **Line mimic**
   stopped station red, root cause on the utilities rail
4. **Live agent trace**
   tool calls, per-claim verdicts, provenance chips
5. **Verified root cause**
   cause, asset, confidence, ranked hypotheses

Layout logic

Left: where. Middle: how the agents got there. Right: what to do, and what a human must approve.

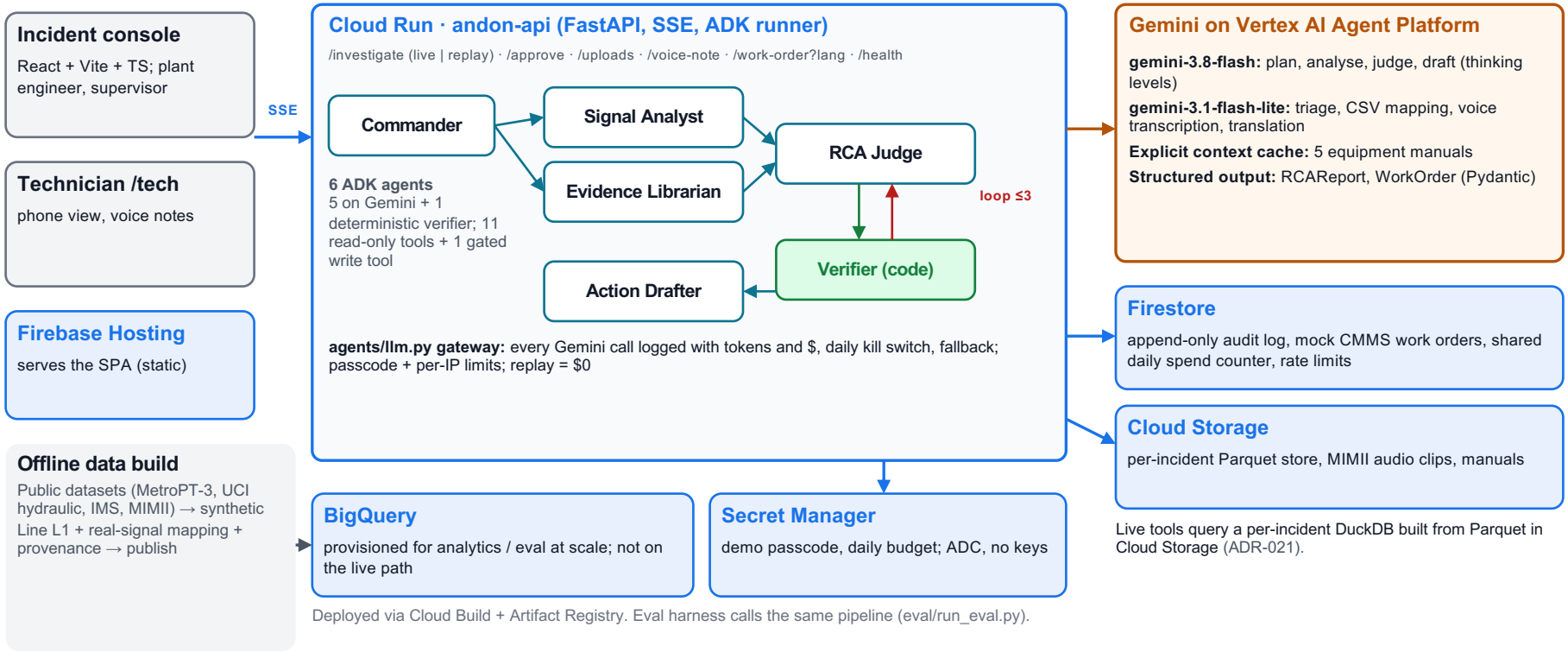


Technician view /tech (phone)

One card at a time: cause → safety checklist → approve, voice note

Architecture diagram

Google Cloud, asia-south1 · Gemini on the global endpoint



Technologies used in the solution

GOOGLE CLOUD

Gemini 3.8 Flash	Commander, Signal Analyst, Librarian, RCA Judge, Action Drafter; thinking LOW → HIGH by role
Gemini 3.1 Flash-Lite	cheap calls: CSV column mapping, voice-note transcription, work-order translation
Vertex AI Agent Platform + ADK	google-adk graph: Sequential, Parallel and Loop agents; google-genai client
Gemini context caching	5 equipment manuals cached once per session, cited by page
Cloud Run	FastAPI API streaming Server-Sent Events, asia-south1
Firebase Hosting	React incident console and /tech technician view
Firestore	append-only audit, mock CMMS work orders, shared budget counter, rate limits
Cloud Storage	per-incident Parquet store, machine audio, manuals
BigQuery	provisioned for analytics / eval at scale; live tools use DuckDB from GCS Parquet (ADR-021)
Secret Manager	passcode and daily budget; service accounts via ADC, no keys

GEMINI CAPABILITIES USED

- Agentic planning + loop revision
- Function calling: 11 read-only tools, 1 gated write
- Audio understanding: listen() on MIMII clips
- Long context + explicit caching
- Structured output (response_schema)
- Thinking levels + cost routing

OPEN-SOURCE STACK

- Python 3.12 · uv
- FastAPI · SSE
- Pydantic v2
- DuckDB
- React · Vite
- TypeScript · Tailwind
- pytest · ruff
- Vitest

Tests run offline with recorded Gemini fixtures; 700+ tests (README).

Estimated implementation cost

measured, then a labelled assumption

$0.107

Gemini cost per diagnosis

S1 mean, S-set; $0.109 R-set (final_v2)

≈ $107

all Gemini spend to build

incl. ≈ $99.8 on evaluation (ledger, Oct 17)

$0

per replay

recorded runs never call a model

WHERE A DIAGNOSIS'S MONEY GOES (77 S1 runs, eval/results/final_v2/cost.json)



Voice note → shift note	≈ $0.0008 per 21 s clip (ADR-029)
Work-order translation	$0.0020–0.0024 per order, cached (ADR-030)
CSV upload investigation	$0.093 live run (ADR-028)
Guard rails	$8/day kill switch; $300 total budget to Dec 4

ASSUMPTION · not measured

Gemini cost for one plant

If a plant runs 1,000 diagnoses a month:

1,000 × $0.107 ≈ $107 / month

Hosting runs on serverless services that scale to zero (Cloud Run, Firebase Hosting, Firestore); our budget sets aside $35 for infrastructure through Dec 4 (allocation, not a bill).

Main levers: replay and caching cost $0; Flash-Lite for cheap steps; the Pro judge was measured worse and dearer ($0.16 vs $0.11 per run) and dropped.

BUILD BUDGET (ledger, DEVELOPMENT.md §7)

$300 total to Dec 4

≈ $107 Gemini spent · ≈ $190 left for judging window and finals

Business value & ROI

one sourced number, the rest labelled assumption

$2.3M

per hour of downtime

Siemens 2024, p.2 (sourced)

→

$38,333

per stopped minute

= $2.3M ÷ 60 (arithmetic)

→

20 min

saved per month, per plant

ASSUMPTION: 10 diagnosed stops × 2 min faster restart; plant-level rate applied

→

≈ $0.77M

value at stake / month

= 20 × $38,333 (assumption-based)

Against a cost of

10 × $0.107 ≈ $1.07 / month

One saved minute of downtime pays for about 358,000 diagnoses ($38,333 ÷ $0.107).

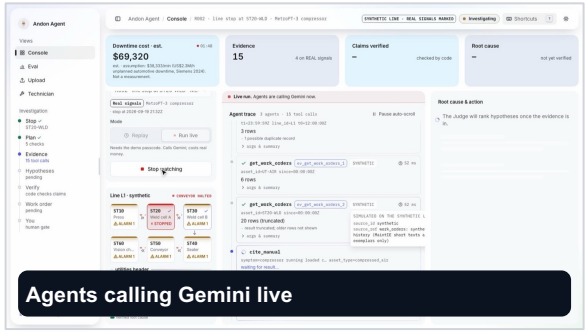
Value we did not price

- Safety: LOTO steps verbatim, every step ticked before approval
- Knowledge capture: every claim, tool call and decision in an audit log
- JAPAC workforce: work orders in Hindi, Tamil, Bahasa Indonesia, Vietnamese

Honest limit: we have not measured time-to-restart in a real plant. The stop rate and minutes saved are illustrative assumptions; a pilot would measure them. Model cost is measured (final_v2).

Snapshots of the prototype

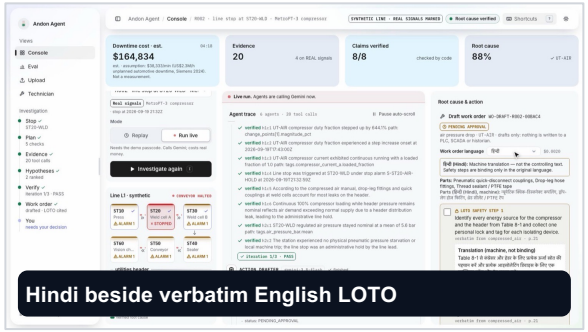
deployed app · live run, 17 Oct 2026



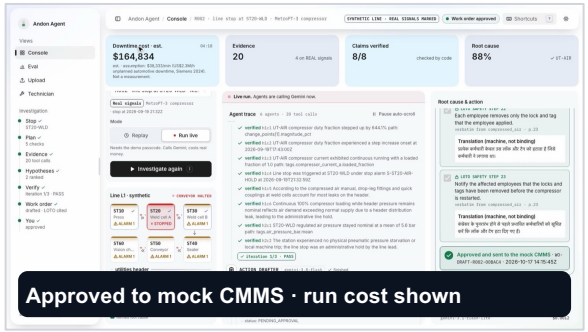
Agents calling Gemini live



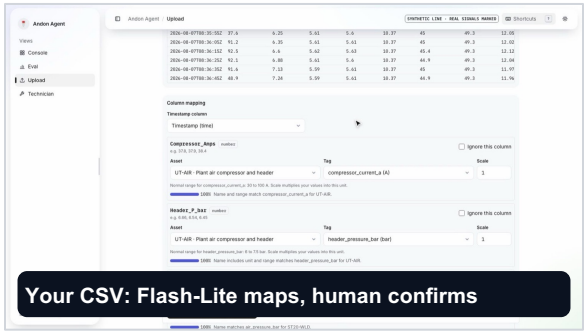
Evidence drawer: real MetroPT-3 window



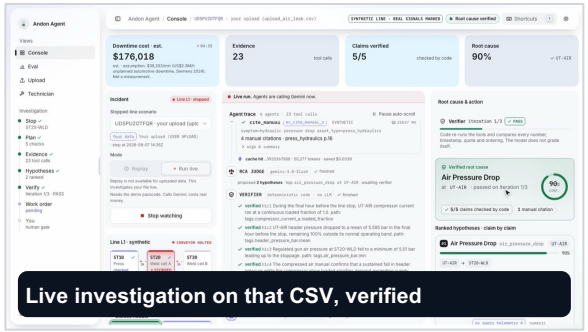
Hindi beside verbatim English LOTO



Approved to mock CMMS · run cost shown



Your CSV: Flash-Lite maps, human confirms



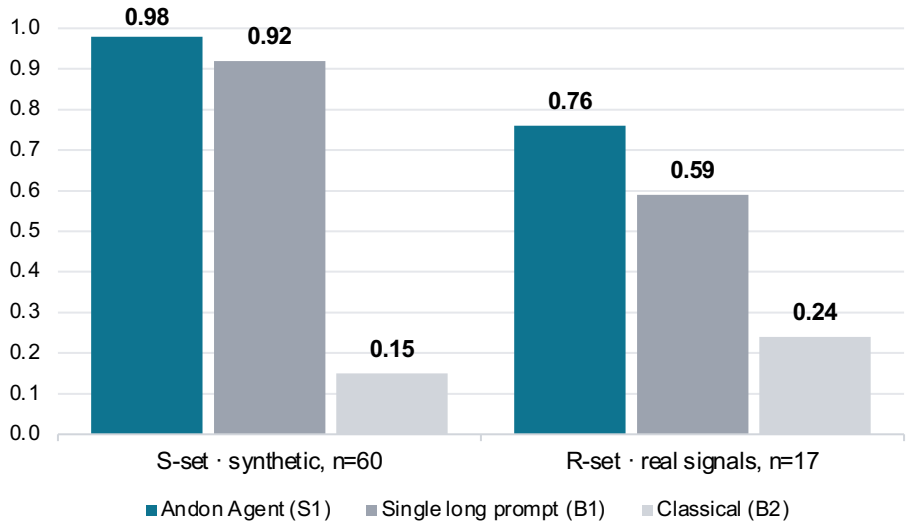
Live investigation on that CSV, verified

Live run, 17 Oct 2026. R002: 20 tool calls, 8/8 claims verified by code, 88% air pressure drop at UT-AIR, 19 LOTO steps, approved; Gemini cost $0.1487. Upload (upload_air_leak.csv): 23 tool calls, 5/5 verified, 90% at UT-AIR.

Prototype performance report / benchmarking

eval run final_v2

Top-1 root cause (right cause class and asset)



95% Wilson CIs: S1 [0.91, 1.00] / [0.53, 0.90] · B1 [0.82, 0.96] / [0.36, 0.78] · B2 [0.08, 0.26] / [0.10, 0.47]. S2, same pipeline with the verifier always passing: 1.00 / 0.76. Failed runs count as misses.

GROUNDING · claims independent code could not back (S + R)

1 / 586 0.2% vs **54 / 645** 8.4%

Andon Agent vs single long prompt with the same evidence. No-verifier ablation (S2): 4 / 609. Grounding comes from evidence-id citations plus the verifier; the verifier's own value is the stress test below.

VERIFIER STRESS TEST · offline, $0 (ADR-032)

97.1% caught [96.4, 97.7] vs **0%** with it off

2,778 faults injected into real S1 reports: wrong numbers, timestamps, misquotes, reversed cause and effect, false premises. Weakest: sign flips 92.6%.

≈ $0.11 per diagnosis · **81 s** median (S-set)

Honest reading: top-1 vs the single prompt is not significant (exact McNemar p = 0.219 S-set, 0.375 R-set). **The win is grounding.**

Going further: fresh-data test and post-hoc rerun

reported separately from the headline

POST-HOC

Measured after two changes made following the final_v2 eval: a judge rule (R002 was unstable across repeats, ADR-026) and an end-anchored short-shift detector (ADR-033). Run eval/results/posthoc_v4.

0.97	0.82	0.50	0 / 648
S-set top-1, n=60	**R-set top-1, n=17**	**C-set top-1, n=10**	**unsupported claims**
[0.89, 0.99]	[0.59, 0.94]	[0.24, 0.76]	S + R + C

Pre-registered test on fresh data (C-set)

Before running, we fixed 10 new scenarios from unused segments of the real sources and a pass bar (top-1 ≥ 0.6). Result: **S1 0.40 vs 0.20 for the single prompt,** with 0 / 80 unsupported claims vs 6 / 91 on data it had never seen. Below our own 0.6 bar, so we kept going (rerun above).

What we are fixing next

Short real hydraulic faults (R009–R011; C-set 0 / 5). A new detector already reports the shift (7 / 10 on a synthetic dev set, 0 false positives in 8,000+ healthy windows); next, the judge must learn to trust a 0.3–6% change over minutes.

OTHER CHECKS (final eval)

Held-out S051–S060: 10 / 10. · Model choice measured, not assumed: Gemini 3.8 Flash beat 3.1 Pro as judge (S 0.98 vs 0.90, R 0.76 vs 0.71) at two-thirds of the cost per run ($0.11 vs $0.16).

Responsible AI & safety

enforced in code and tests

1 No physical writes

No tool can write to a PLC, SCADA or historian. The only write tool drafts PENDING_APPROVAL work orders in a mock CMMS; a named human approves.

2 Safety text is verbatim

Lockout/tagout steps are exact manual quotes with page numbers, inserted by code (ADR-025); structure follows OSHA 29 CFR 1910.147. Translations sit beside, never replace.

3 Every claim checked by code

Numbers, timestamps, quotes, causal order and citation premises are re-run. Unverifiable claims are struck through, never dropped.

4 Untrusted text stays data

Shift notes, CSV uploads, voice notes and stop text are wrapped as untrusted; voice notes never reach an agent and cannot change a work order.

5 Honest data

Public datasets under their licences; real signals mapped onto a synthetic line, every real segment carries provenance; the UI shows SYNTHETIC LINE · REAL SIGNALS MARKED.

6 Access, cost, audit

Live runs need a passcode, per-IP limits, $8/day kill switch; append-only audit of every model call and human action; secrets in Secret Manager.

Residency mode (designed, not deployed): pin Gemini 3.5 Flash to asia-south1 / asia-southeast1 for Indian and Singapore plants. Equipment manuals are fictional, written for this demo.

Additional details / future development

NEXT

Close the known gap

- Teach the judge to trust the short-shift detector (short real hydraulic faults)
- Historian / OPC UA / MQTT adapters behind the same TelemetryStore interface (CSV upload proves the pattern)
- Residency mode live in asia-south1 / asia-southeast1

SCALE

From one line to a plant

- Dataproc ingestion for plant-wide history
- Vector search once work orders pass ~1M (SQL by asset and time today)
- Gemma 4 at the edge for plants that cannot send data out

PROVE

Pilot and finale

- Pilot with a plant to measure time-to-restart (today an assumption)
- More real-signal scenarios graded on source labels
- If selected, show it live at the Singapore finale, Dec 4

Known limits today: the line is synthetic (real signals mapped onto it); the real-signal set is small (n = 17); translations are machine output, labelled as not binding; uploads are held per instance, so upload mode runs one instance.

Built fresh in the hackathon window: dated public commits; 33 architecture decisions logged (docs/DECISIONS.md); 700+ offline tests; every number in this deck traces to a committed results file.

Links

1	**GitHub public repository**	https://github.com/Likhith-BlueLotus/andon-agent
2	**Demo video (3:29)**	<YouTube link — added at upload>
3	**Final product (live app)**	https://andon-agent-b9c77.web.app
4	**API health**	https://andon-api-359520289475.asia-south1.run.app/health

60-second judge path: open the live app → pick R002 · MetroPT-3 compressor (replay) → Investigate stop → click an evidence chip → tick the LOTO steps and approve. Replay S014 shows the verifier rejecting a reversed cause-and-effect claim. Replays need no login; live investigations call Gemini and need the demo passcode.

Theme: **Manufacturing: Intelligent Operations & Industrial Efficiency** · code MIT · data sources and licences: data/SOURCES.md

Built like a product team

Two full-stack builders worked through nine role lenses. Each lens is backed by a role-specialised AI review agent (.claude/agents/), and a sceptical judge agent scored us against the rubric.

Principal Engineer / Architect

Architecture, the three contracts, 33 ADRs, cost and latency budgets, scope cuts

Staff Engineer · agents

ADK graph, prompts, tool wiring, model routing and fallback, verifier loop, context cache

Senior Backend Engineer

FastAPI + SSE, Firestore audit and mock CMMS, replay, passcode, rate limits, kill switch

Senior Data & ML Engineer

Dataset fetch and provenance, synthetic line, real-signal mapping, fault injector, signal maths

SRE / DevOps

GCP bootstrap, IAM, Cloud Run, Firebase Hosting, Secret Manager, budgets, deploy scripts

UX / UI Engineer

Incident console, trace and provenance chips, evidence drawer, technician view, accessibility

QA & Eval Engineer

Offline tests with recorded fixtures, eval harness and baselines, Wilson CIs, McNemar, stress test

Security & Responsible-AI

Read-only reviews before each deploy: IAM, secrets, prompt injection, safety rules, licences

Product & Pitch Lead / BA

Personas, demo script, deck, judge Q&A, ROI assumptions, sourcing of every number

Ownership: Person A: agents, backend, infra, eval harness · Person B: data (S-set, R-set), frontend, deck and video.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

